CO2 - AT 1

Analytical Problem Solving

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COURSE : Data Structures

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np)

(1) Queue size = 5
Front = -1
Rear = -1

	Operation	Queue(0-4)	Front	Rear
1.	Enqueue (20)	20 - - - -	0	0
2.	Enqueue (10)	20 10 - - -	0	1
3.	Enqueue (30)	20 10 30 - -	0	2
4.	Dequeue (20)	_ 10 30 - -	1	2
5.	Enqueue (80)	- 10 30 80 -	1	3
6.	Dequeue (10)	- - 30 80 -	2	3
7.	Enqueue (90)	- - 30 80 90	2	4
8.	Enqueue (100)	100 - 30 80 90	2	0
9.	Dequeue (30)	100 - - 80 90	3	0
10.	Dequeue (80)	100 - - - 90	4	0
11.	Enqueue (60)	100 60 - - 90	4	1
12.	Enqueue (90)	100 60 90 - 90	4	2

Final Queue

Index	0	1	2	3	4
Value	100	60	90	-	90

Final Positions

Front = 4

Rear = 2

Front element = 90

Rear element = 90

(2) Binary Search Element

Elements

66 , 111 , 212, 323, 432, 543, 643, 649, 777, 8

Index	0	1	2	3	4	5	6	7	8	9
Value	66	111	212	323	432	543	643	649	777	88

Binary Search Algorithm

binarysearch (arr, key)
low = 0
high = n-1

```
while (low <= high)
{ mid = (low + high)/2
  if (arr[mid] == key)
      return mid ;
  else if (key < arr[mid])
      high = mid -1
  else
      low = mid +1
 }
return -1
```

Case 1: Key = 111

$$mid = (low + high)/2$$
$$= (0 + 9)/2$$
$$= 4.5 \simeq 4$$

$$mid \neq Key$$
$$432 > 111$$

So, $high = mid - 1$

$$mid = (0 + 3)/2$$
$$= 1$$

mid = Key

111 = 111

$\therefore$ Element found at Index 1.

Case 2: Key = 888

mid = 4

middle element $\neq$ key

$432 < 888$

So, low = mid + 1

mid = $(5 + 9)/2$

$= 7$

middle element $\neq$ key

$649 < 888$

So, low = mid + 1

mid = $(8+9)/2$

$= 8.5 \simeq 8$

middle element $\neq$ key

$777 < 888$

So, low = mid + 1

mid = 9

$\therefore$ Element found at index 9.

Case 3: Key = 999

mid = 4

middle element $\neq$ key

432 < 999

So, low = mid + 1

mid = 7

middle element $\neq$ key

649 < 999

So, low = mid + 1

mid = 8

middle element $\neq$ key

777 < 999

So, low = mid + 1

mid = 9

middle $\neq$ key

888 < 999

So, Element not found in the array.